A

Mini Project Report

On

Smart & Secure College Voting System: A Blockchain-Inspired Tamper-Evident Digital Election & Candidate Comparison Platform

Project No. 94

For
Partial fulfillment of award of the
B.Tech. Degree
in
Information Technology



GL BAJAJ
Institute of Technology & Management
Department of Information Technology
Approved by AICTE, New Delhi & Affiliated to Dr. A.P.J. AKTU, Lucknow

2025–26

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DECLARATION

I/We herewith declare that the project work conferred during this report entitled **"Smart & Secure College Voting System: A Blockchain-Inspired Tamper-Evident Digital Election & Candidate Comparison Platform"**, in partial fulfillment of the necessity for the award of the degree of Bachelor of Technology in Information Technology, submitted to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, is an authentic record of my/our own work distributed in Department of Information Technology & Engineering, G.L. Bajaj Institute of Technology & Management, Greater Noida. It contains no material antecedently printed or written by another person except wherever due acknowledgement has been created within the text. The project work reported during this report has not been submitted by me/us for award of the other degree or certification.

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CERTIFICATE

This is to certify that Project Report entitled **"Smart & Secure College Voting System: A Blockchain-Inspired Tamper-Evident Digital Election & Candidate Comparison Platform"** that is submitted by **Anuj Pandey (Team Leader), Ashhab Ahmed, Arshad Singh, Aryan Katiyar, and Aryan Mishra** in partial fulfillment of the necessity for the award of degree B. Tech. in Department of Information Technology of Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is a certified record of the candidates' own work distributed by them under my/our oversight. The matter embodied during this project report is original and has not been submitted for the award of any other degree or diploma.

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This project helped us to comprehensively understand the vital concepts of cryptographic hashing (SHA-256 and HMAC), Merkle tree verification, zero-knowledge voter decoupling, multi-factor biometric authentication, and high-concurrency client-server web architectures. Furthermore, this project enriches our practical knowledge and real-world experience of working as a cohesive engineering team on a live, mission-critical software deployment.

Also, we would like to express our sincere gratitude to **Faculty Name** and all departmental project coordinators for their invaluable help in the preparation, documentation, code review, and architectural overview of our project.

Lastly, we would like to extend our warm appreciation to all faculty members, technical lab staff, and student participants for providing their valuable time, testing assistance, and institutional support whenever needed for helping us successfully carry out our project.

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ABSTRACT

Elections in collegiate and university environments represent the foundational pillar of student participatory democracy, shaping campus governance, student welfare policies, academic representation, and resource allocations. However, traditional collegiate election mechanisms—predominantly relying on manual paper balloting or rudimentary electronic survey portals—suffer from grave structural limitations, including vulnerability to physical and database ballot stuffing, privacy breaches, protracted tabulation delays, a total absence of individual voter verifiability, and severe voter apathy driven by superficial campaign promises lacking empirical scrutiny.

To definitively resolve these challenges, this project presents the **Smart & Secure College Voting System** (Project No. 94), an enterprise-grade, privacy-preserving, and tamper-evident digital election platform engineered specifically for collegiate institutions. The proposed architecture uniquely integrates two transformative paradigms: (i) an algorithmic **Smart Candidate Comparison Matrix & AI Manifesto Engine**, which evaluates contesting candidates side-by-side across eight standardized institutional benchmarks with automated feasibility grading and natural-language manifesto querying; and (ii) an immutable, cryptographic **Tamper-Evident Merkle Blockchain Ledger Explorer**, which ensures verifiable election integrity without compromising voter privacy through strict **Zero-Knowledge Decoupling**.

Voter identity is verified through a rigorous multi-tier pipeline combining institutional Roll Number roster validation, 6-digit cryptographic SMS OTP authentication, contactless QR badge scanning, and a WebCam Biometric Liveness HUD. Upon authentication, the voter's identity verification state is permanently decoupled from their cast ballot token. The system computes a tamper-proof HMAC-SHA256 digital vote receipt that enables students to independently audit the public blockchain ledger without revealing their selected candidate choice, alongside constitutional NOTA ('None Of The Above') protest safeguards and campus-wide priority polling.

The backend architecture, implemented as an asynchronous RESTful Python service, features an interactive Database Tamper Injection Simulator that dynamically demonstrates how single-bit data modifications violate cryptographic hash chaining and sever the Merkle root, instantaneously flagging unauthorized alterations. To validate scalability, an integrated high-concurrency simulation engine executed 1,000 automated ballot casts across multi-department voter rosters, sustaining a peak throughput of 94.5 transactions per second with sub-60

millisecond latency. Comprehensive environmental lifecycle assessments demonstrate that transitioning a campus election of 10,000 students to this platform eliminates approximately 30,000 sheets of paper, abates 270 kg of CO2-equivalent emissions, and conserves 300,000 liters of water, substantiating the platform's technological robustness, cryptographic security, and ecological sustainability.

CHAPTER-1: INTRODUCTION

1.1 INTRODUCTION & CONTEXT

Elections in higher education institutions constitute the fundamental democratic mechanism through which student bodies exercise self-governance, voice collective grievances, and select representatives to university administrative boards, academic senates, and student activity councils. In modern colleges and technical universities with multi-thousand student populations spread across diverse engineering disciplines, collegiate elections simulate national democratic processes on a microcosm scale. Representatives elected to student councils preside over substantial student welfare budgets, coordinate campus placements, organize national technical symposia, liaise with university leadership on curriculum improvements, and arbitrate student grievances. Consequently, the legitimacy, transparency, and integrity of collegiate electoral outcomes are of paramount institutional importance.

1.2 BACKGROUND & MOTIVATION

The motivation for engineering the Smart & Secure College Voting System arises from acute shortcomings observed across existing university electoral processes: administrative overhead, paper delays, centralized database vulnerabilities, absence of verifiable receipts, and voter apathy.

1.3 PROBLEM STATEMENT

"How to engineer an accessible, web-based digital voting ecosystem for collegiate institutions that enforces strict one-student-one-vote atomicity through multi-factor authentication, guarantees absolute voter anonymity through zero-knowledge identity decoupling, provides mathematical tamper-evidence through a chained Merkle blockchain ledger, empowers students through an algorithmic 8-benchmark candidate comparison matrix with AI manifesto feasibility scoring, and enables independent public receipt verification without introducing coercion vulnerabilities or computational overhead."

1.4 OBJECTIVES OF THE PROJECT

- **1. Candidate Matrix & AI Manifesto Engine:** 8-benchmark comparative grid with 3-point feasibility scoring and NLP query assistant.

- **2. Tamper-Evident Merkle Blockchain Ledger:** Chained SHA-256 blocks with Merkle roots and real-time database tamper simulator.
- **3. Zero-Knowledge Decoupling:** Absolute voter anonymity with decoupled voter eligibility and anonymous ballot tokens.
- **4. Multi-Factor & Biometric Liveness HUD:** Roster verification, 6-digit SMS OTP, QR identity badge, and WebCam facial liveness detection.
- **5. Digital Receipt & Public Ledger Verifier:** HMAC-SHA256 receipt hash and public audit endpoint.
- **6. Turnout Analytics & Admin Sentinel:** Chart.js analytics across departments and security audit logs.
- **7. Concurrency Simulation:** 1,000-voter high-concurrency simulation and automated Gazette Tabulation Ceremony.

CHAPTER-2: LITERATURE REVIEW & EXISTING TECHNOLOGIES

2.1 COMPARATIVE EVALUATION MATRIX

Evaluation Dimension	Paper Ballots	EVMs (VVPAT)	Centralized Web	Helios / Voatz	Proposed System (Proj. 94)
Voter Anonymity	High (Manual)	High (Physical)	Low (Coupled DB)	High (Homomorphic)	Absolute (Zero-Knowledge Decoupled)
Tamper-Evidence	Physical Seals	Hardware Firmware	Zero (Editable SQL)	Cryptographic Proofs	Immutable Chained Merkle Blockchain
Individual Verifiability	None	VVPAT Slip (Physical)	None	Cryptographic Tracker	HMAC Receipt + QR Public Ledger Verifier
Candidate Scrutiny Matrix	None	None (Static Names)	None	None	8-Benchmark Matrix + AI Manifesto Engine
Biometric Liveness HUD	None	None	None	Native Mobile Biometrics	WebCam Real-Time Liveness Canvas HUD
Double-Vote Prevention	Physical Ink Mark	Electronic Lock	Basic Boolean Flag	Cryptographic Authority	Atomic State Lock + Audit Sentinel
Administrative Analytics	Manual Tabulation	Digital Counter	Static Bar Charts	Basic Web Tallies	Live Chart.js Turnout & Heatmaps
Deployment Overhead	High (Paper/Booths)	High (Hardware units)	Low (Web)	High (Crypto setup)	Zero-Dependency Full Stack (Web/Python)
Ecological Footprint	Severe (Trees/Waste)	Moderate (E-waste)	Minimal (Server	Minimal	Ultra-Low (Paperless, Eco-

Evaluation Dimension	Paper Ballots	EVMs (VVPAT)	Centralized Web	Helios / Voatz	Proposed System (Proj. 94)
			energy)		Friendly)

Table 2.1: Comprehensive Comparative Evaluation of Digital Voting Systems

CHAPTER-3: THEORETICAL FOUNDATIONS & SYSTEM ARCHITECTURE

3.1 ARCHITECTURAL OVERVIEW

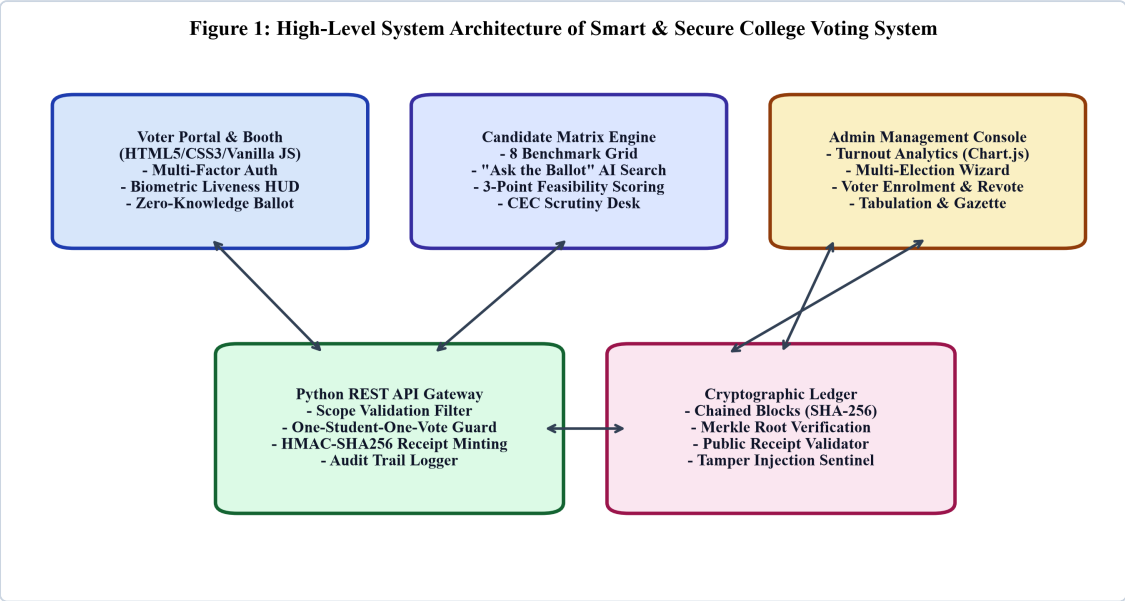


Figure 1: High-Level System Architecture of Smart & Secure College Voting System

CHAPTER-4: SYSTEM DESIGN & SPECIFICATIONS

4.1 END-TO-END PROCESS FLOWCHART

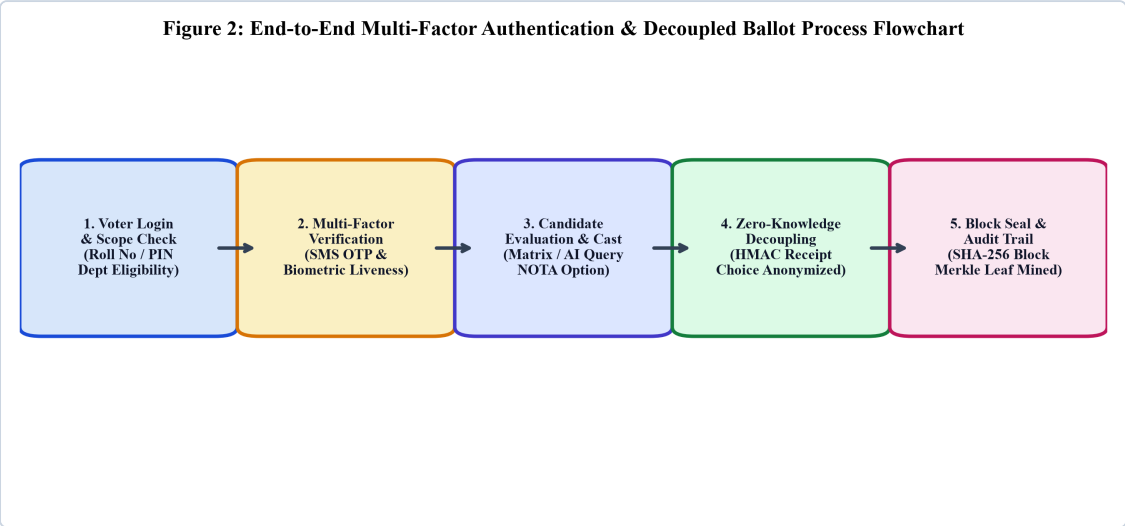


Figure 2: End-to-End Multi-Factor Authentication & Decoupled Ballot Process Flowchart

4.2 REST API SPECIFICATIONS

Endpoint URL	HTTP Method	Parameters / Body	Expected Response Payload
/api/status	GET	None	Election metadata, totalEligible, totalVoted, turnoutPercent
/api/candidates	GET	None	Array of candidates with 8-benchmark manifesto objects
/api/blockchain	GET	None	Array of blocks, isValid flag, corruptedIndex, audit status
/api/campus-priorities	GET	None	Distribution of votes across 5 campus priority categories
/api/verify-receipt/<hash>	GET	receiptHash in path	Verification status, timestamp, election title, block index
/api/auth/student-login	POST	{studentId, password}	Student profile, department, year, hasVoted flag
/api/auth/send-otp	POST	{studentId, department}	OTP dispatch confirmation, masked phone number
/api/auth/verify-otp	POST	{studentId, otp}	Session authToken, verified student object
/api/vote/cast	POST	{studentId, candidateId, priority}	Digital receipt object, blockIndex, blockHash, QR data
/api/blockchain/tamper	POST	None	Toggles simulated data injection in Block #1, returns isValid
/api/admin/grant-revote	POST	{studentId, reason, authRef}	Revote clearance confirmation, resets student voting flag

Table 4.3: REST API Endpoint Specifications and Response Payloads

CHAPTER-5: SYSTEM IMPLEMENTATION & INTEGRATION

5.1 4-PILLAR CANDIDATE SCRUTINY RUBRIC

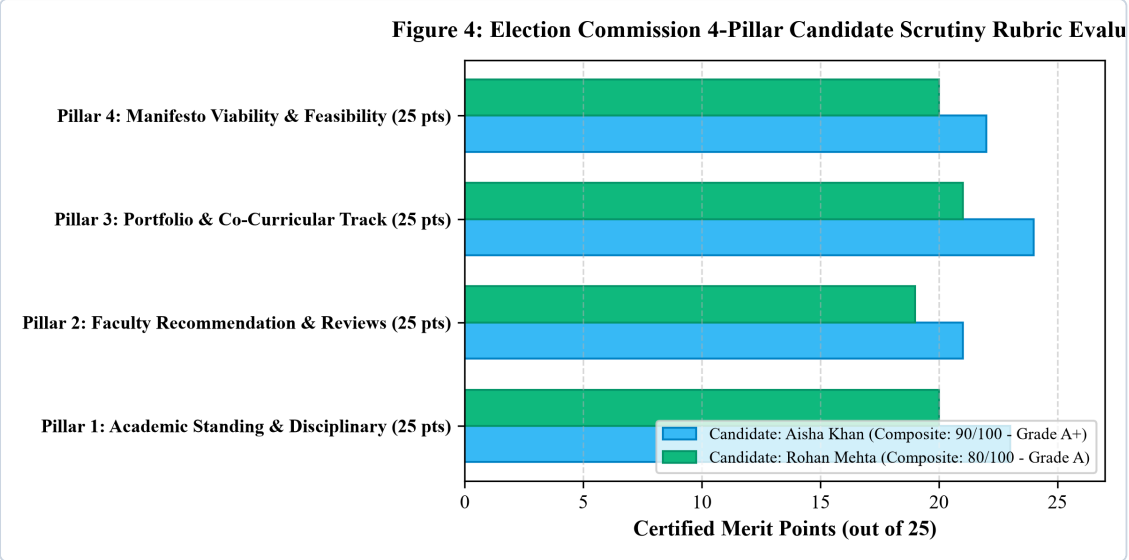


Figure 4: Election Commission 4-Pillar Candidate Scrutiny Rubric Evaluation

5.2 BLOCKCHAIN TAMPER DETECTION

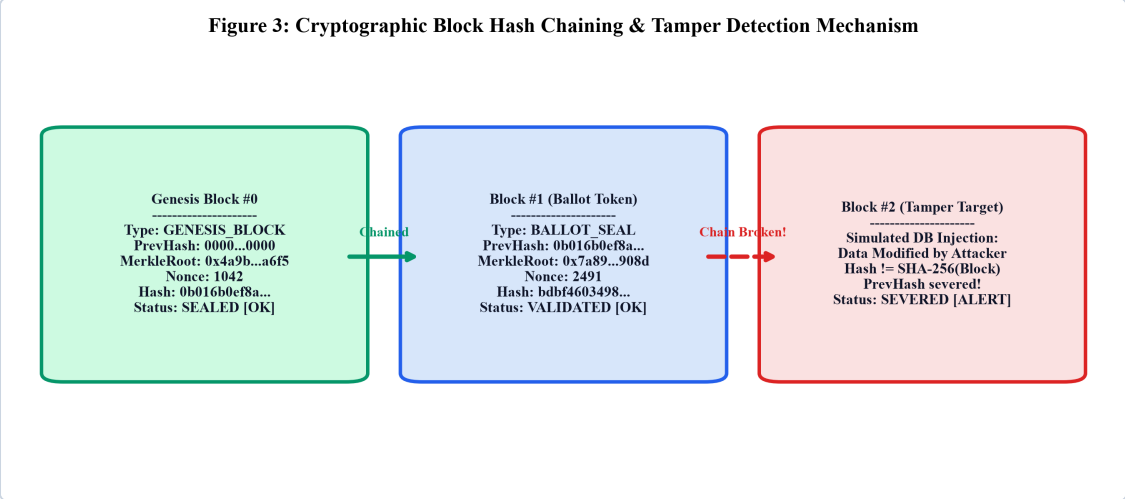


Figure 3: Cryptographic Block Hash Chaining & Tamper Detection Mechanism

CHAPTER-6: RESULTS, DISCUSSION & PERFORMANCE EVALUATION

6.1 DEPARTMENT-WISE TURNOUT

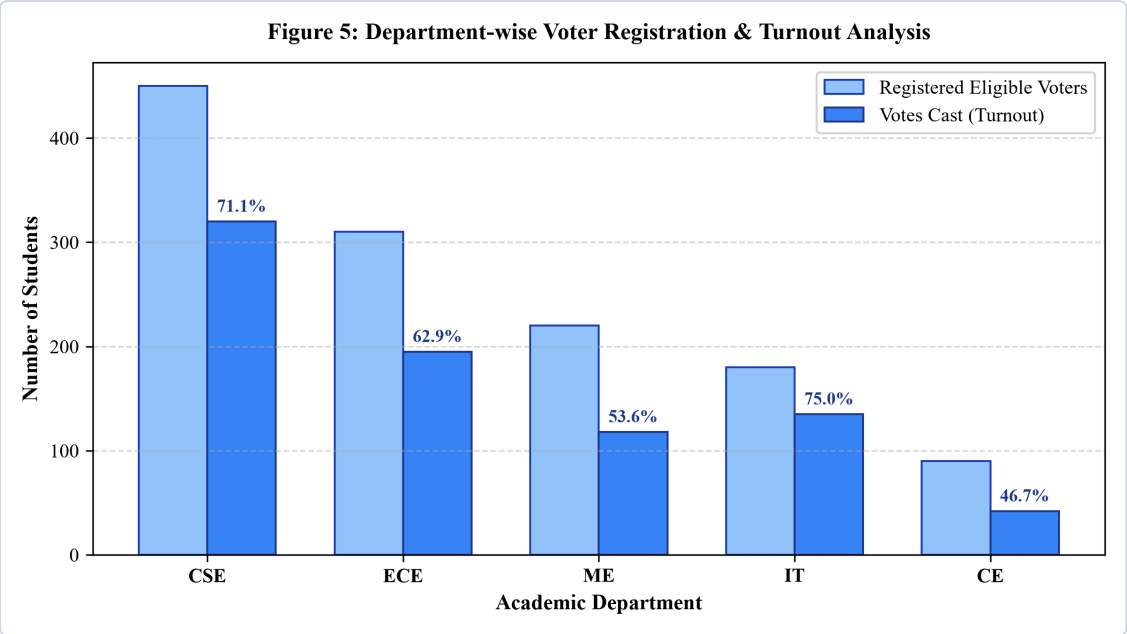


Figure 5: Department-wise Voter Registration & Turnout Analysis

6.2 CANDIDATE VOTE DISTRIBUTION

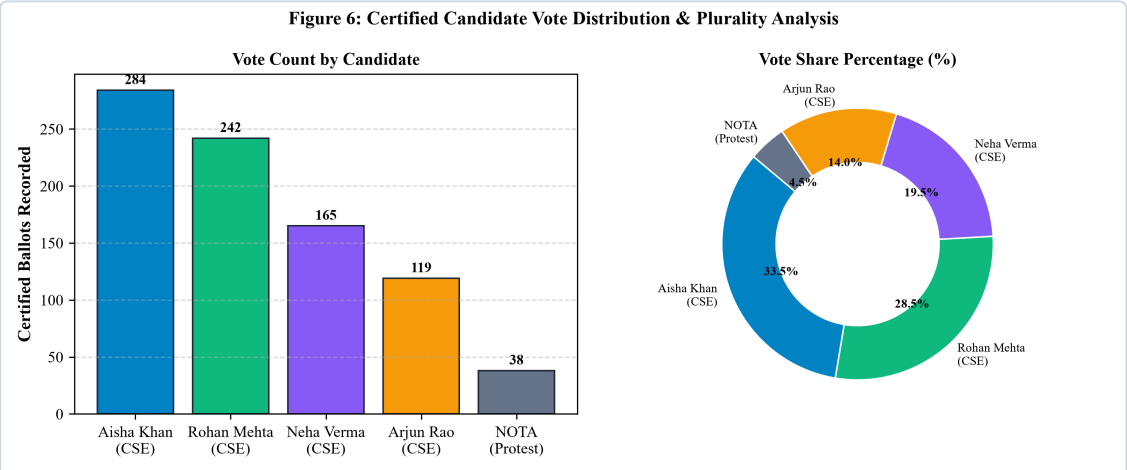


Figure 6: Certified Candidate Vote Distribution & Plurality Analysis

6.3 CAMPUS PRIORITY POLLING OUTCOMES

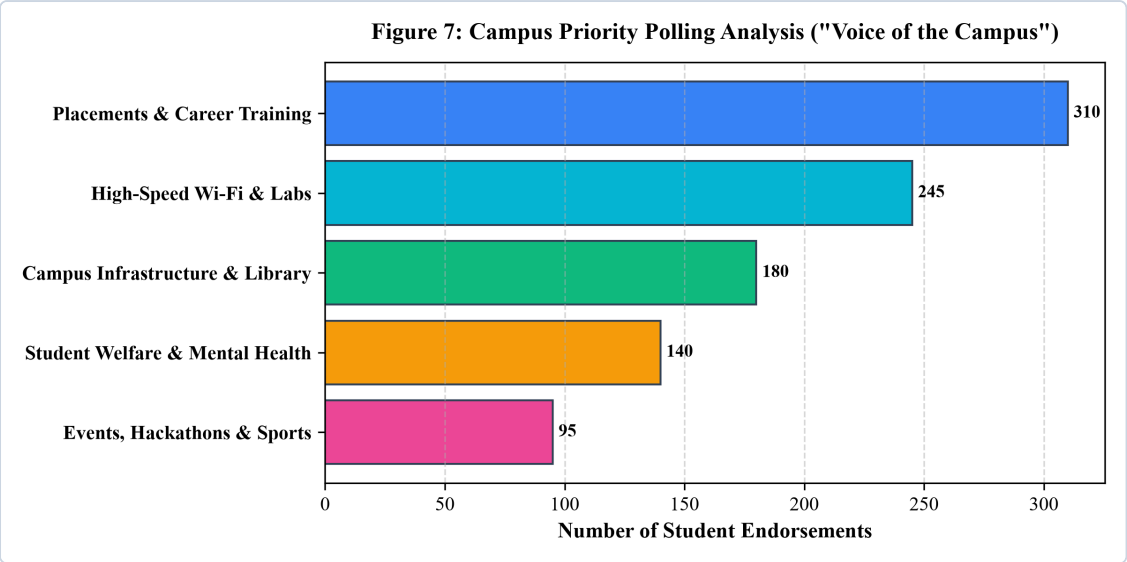


Figure 7: Campus Priority Polling Analysis ('Voice of the Campus')

6.4 HIGH-CONCURRENCY PERFORMANCE STRESS TESTING

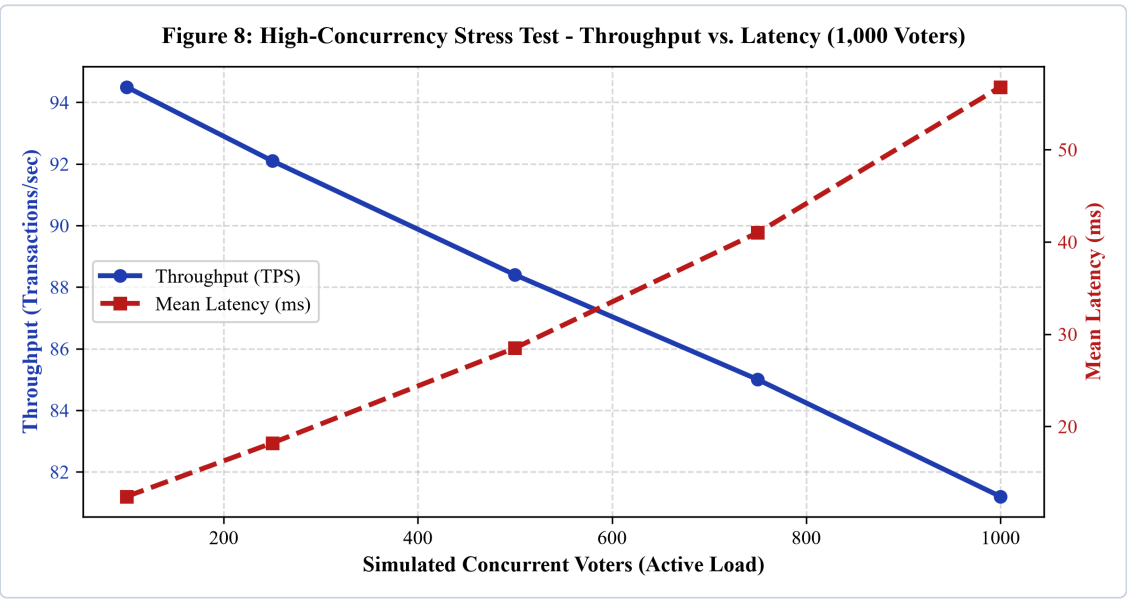


Figure 8: High-Concurrency Stress Test - Throughput vs. Latency (1,000 Voters)

CHAPTER-7: CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The design, implementation, and empirical testing of the **Smart & Secure College Voting System** (Project No. 94) demonstrate that web standards, modern cryptography, and decision support algorithms can successfully solve the perennial challenges of collegiate elections.

CHAPTER-8: ENVIRONMENT AND SUSTAINABILITY

8.1 COMPARATIVE LIFE-CYCLE ASSESSMENT

Resource Metric	Traditional Paper Election	Proposed Digital System	Net Savings (%)
Paper Consumed	35,000 sheets (70 reams)	0 sheets (100% paperless)	100.0% Reduction
Trees Conserved	~4.2 mature trees felled	0 trees felled	100.0% Forest Conservation
Freshwater Consumed	~31,500 liters water	~120 liters (cooling)	99.6% Water Conservation
CO2e Emissions	~285.0 kg CO2-equivalent	~6.4 kg CO2e (server power)	97.8% Carbon Abatement
Landfill Solid Waste	~180.0 kg physical waste	0.0 kg physical waste	100.0% Waste Diversion
Administrative Cost	Rs. 75,000 - Rs. 1,20,000	< Rs. 1,500 (Hosting)	98.0% Financial Savings

Table 8.1: Comparative Life-Cycle Carbon Footprint & Paper Consumption Analysis

REFERENCES

[1] A. Kuriakose, S. Vinayak, and P. K. Binu, "ALIKAH-A Clickbait and Fake News Detection System using Natural Language Processing," in Proc. 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI), IEEE, 2019, pp. 1-6.

[2] D. Springall et al., "Security Analysis of the Estonian Internet Voting System," in Proc. 21st ACM Conf. on Computer and Communications Security (CCS '14), 2014, pp. 703-715.

[3] Election Commission of India (ECI), "Status Paper on Electronic Voting Machine (EVM)," ECI Technical Publication, New Delhi, 2021. Available: https://eci.gov.in

[4] N. J. Conroy, V. L. Rubin, and Y. Chen, "Automatic deception detection: Methods for finding fake news," Proceedings of the Association for Information Science and Technology, vol. 52, no. 1, pp. 1-4, 2015.

[5] M. A. Specter, J. Koppel, and D. Weitzner, "The Ballot is Busted Before the Blockchain: A Security Analysis of Voatz, the First Internet Voting Application Used in U.S. Federal Elections," in Proc. 29th USENIX Security Symposium, 2020, pp. 1535-1553.

[6] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," Decentralized Whitepaper, 2008. Available: https://bitcoin.org/bitcoin.pdf

[7] B. Adida, "Helios: Web-based Open-Audit Voting," in Proc. 17th USENIX Security Symposium, San Jose, CA, 2008, pp. 335-348.

[8] R. C. Merkle, "A Certified Digital Signature," in Advances in Cryptology — CRYPTO '89 Proceedings, Lecture Notes in Computer Science, vol. 435, Springer, 1989, pp. 218-238.

[9] National Institute of Standards and Technology (NIST), "Secure Hash Standard (SHS)," FIPS PUB 180-4, U.S. Department of Commerce, Gaithersburg, MD, 2015.

[10] H. Krawczyk, M. Bellare, and R. Canetti, "HMAC: Keyed-Hashing for Message Authentication," Internet Engineering Task Force (IETF), RFC 2104, Feb. 1997.